

Brain Tumor Detection from MRI Images

Project Objective & Medical Relevance

Objective

Develop an AI system to automatically detect brain tumors from MRI scans with high accuracy and explainability

Why It Matters

- Early detection saves lives
- Reduces radiologist workload
- Faster diagnosis = faster treatment
- **“for aggressive cases diagnosing a Tumor just one month earlier can reduce the risk of mortality 18% to 28%” (Boyne et al., 2025)**

Medical Impact

Brain tumors:

- 700,000+ cases worldwide/year
- Critical for treatment planning

AI can:

- Flag high-risk cases immediately
- Assist in low-resource settings

Dataset Summary



253

Total Images



155

Tumor Cases



98

No Tumor



224×224

Image Size

Dataset Details

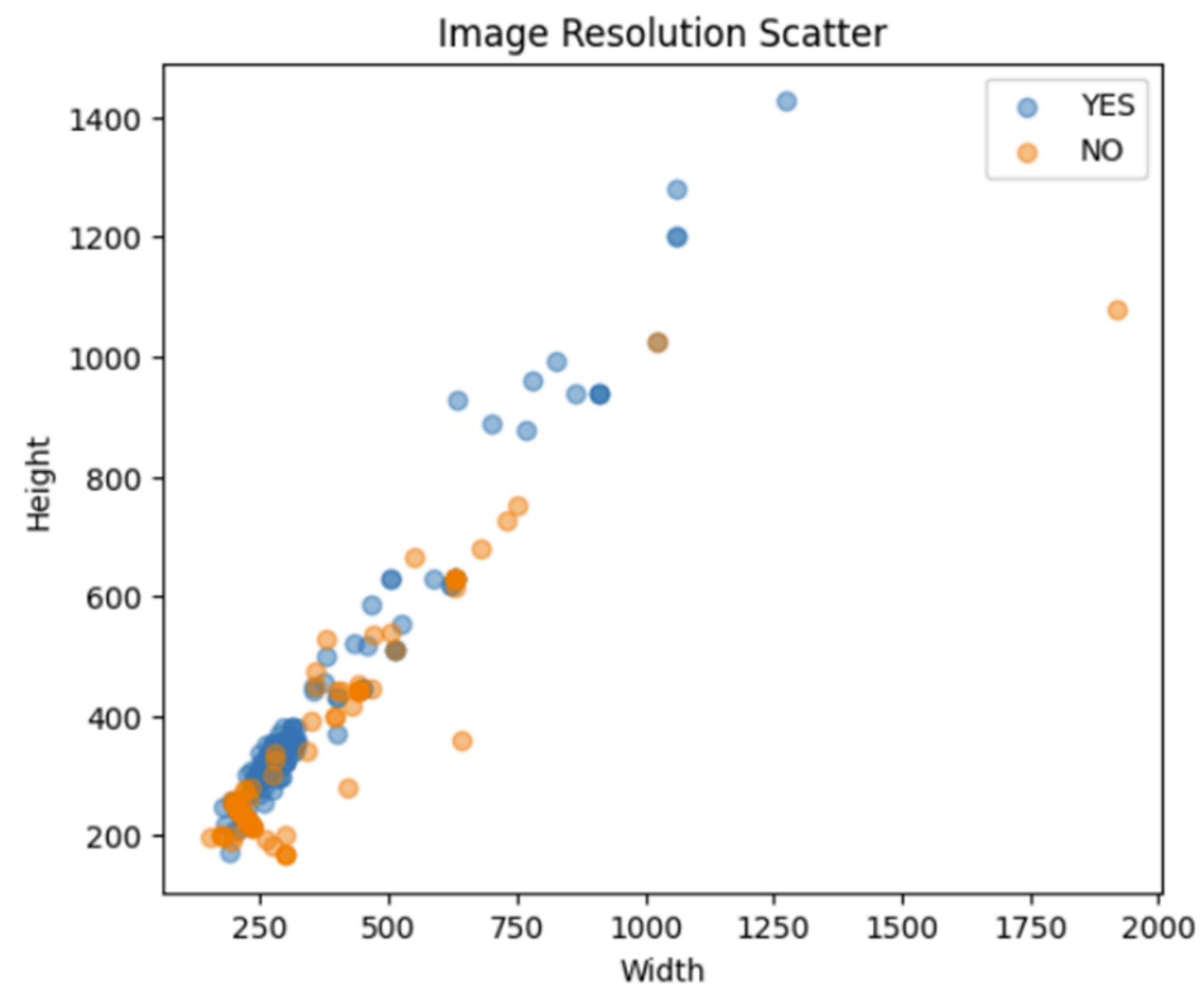
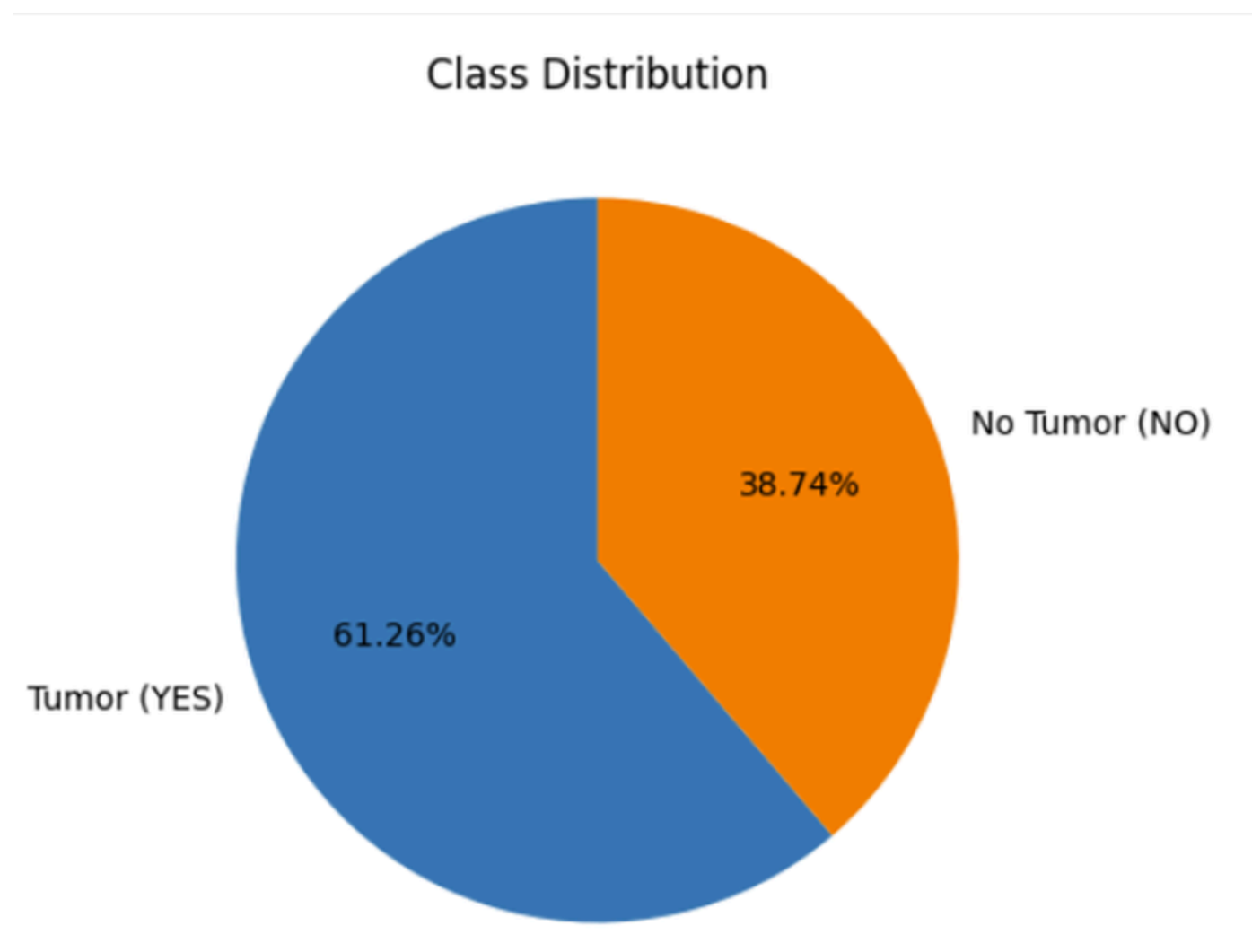
Source: Kaggle Brain MRI Dataset

Structure: yes/ (tumor) and no/ (no tumor) folders

Format: Grayscale MRI stored as 3-channel RGB (R=G=B)

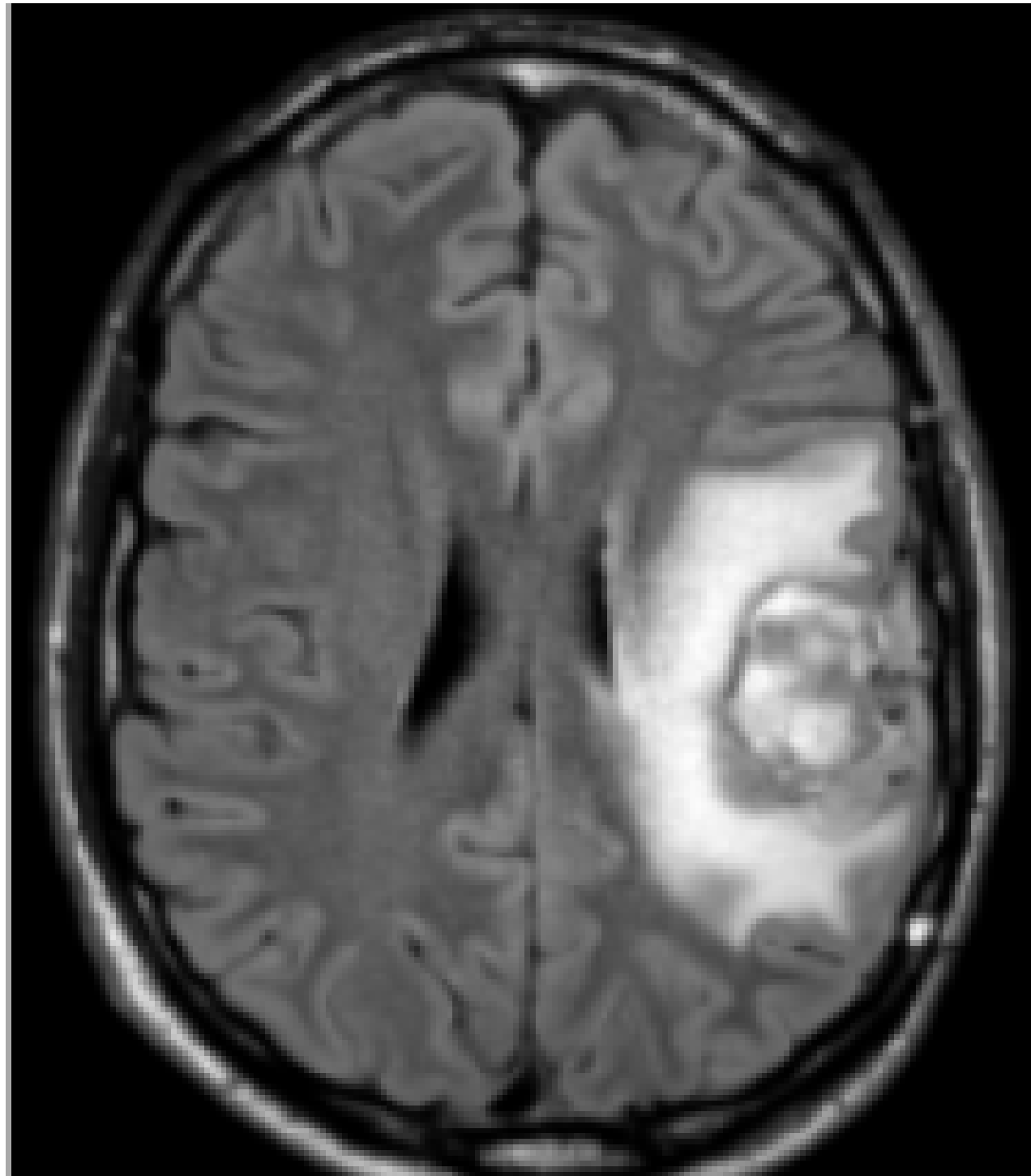
Split: 70% train (177) · 15% validation (38) · 15% test (38)

Class Imbalance: 1.58:1 ratio (tumor:no-tumor) → handled with class weights

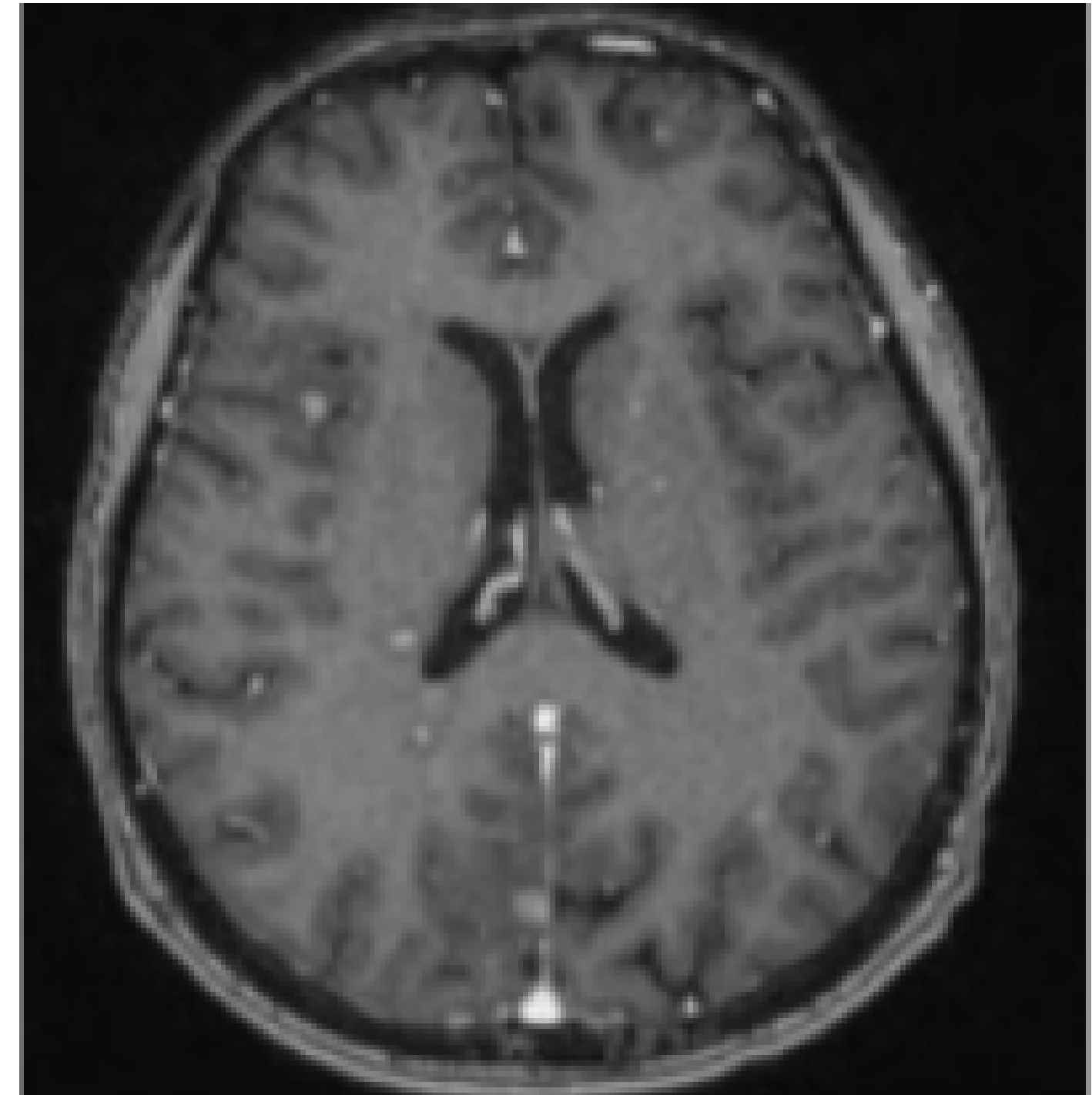


Sample MRI Images from Dataset

Tumor Examples (yes/)



No Tumor Examples (no/)



Key Observation: Tumors appear as bright/white irregular regions

Data Preprocessing Pipeline

1

Data Cleaning

- Verify image validity
- Check for corruption
- Remove duplicates

2

Resize & Normalize

- Resize to 224×224
- Scale to [0, 1]
- ImageNet normalization

3

Data Split

- 70/15/15 split
- Stratified sampling
- Preserve class ratio

4

Augmentation

- Random flips
- Rotation ($\pm 8^\circ$)
- Color jitter



Why These Steps Matter

- Augmentation prevents overfitting on small dataset (177 images) • ImageNet normalization enables transfer learning
- Stratification preserves 1.58:1 class ratio across splits

Model Design & Rationale

Model 1: Custom CNN

- **Architecture:**
 - • 3 convolutional blocks
 - • 32 → 128 gradual feature expansion
 - • MaxPool2d(2X2) ->(224-112-56-28)
 - BatchNorm at every layer—stabilizes training
 - • Classifier: Flatten → 128-2 classes
- **Rationale:**
 - • 3 blocks provide sufficient capacity
 - • AdaptiveAvgPool reduces spatial dimensions efficiently and ensures input size flexibility (at Flattening)
 - • Baseline for comparison

Expected: 75-80% accuracy

Model 2: EfficientNet-B0

Architecture:

- • Pretrained on ImageNet
- • 4M+ parameters
- • Replace final classifier
- • Fine-tune entire model

Rationale:

- • Transfer learning from 1.2M images
- • State-of-art feature extraction
- • Best for small datasets

Expected: 90-95% accuracy

Training Strategy & Hyperparameters

Common Settings

Loss Function: CrossEntropyLoss with class weights

Optimizer: Adam with weight decay (1e-4)

Batch Size: 32 (small dataset)

Early Stopping: Patience = 7 epochs (prevents overfitting)

Custom CNN

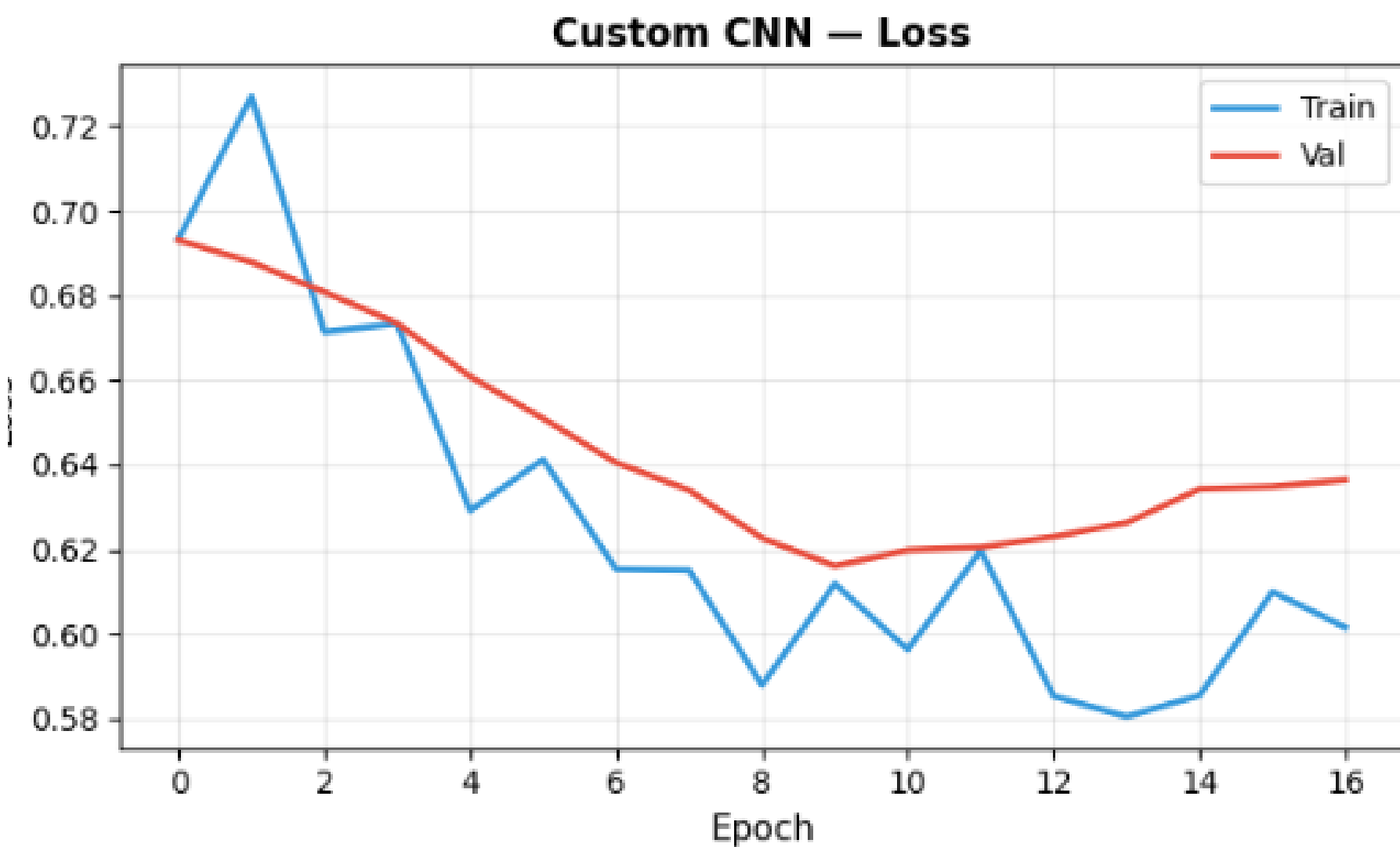
- Learning Rate: 1e-4 (provides stable convergence)
- Dropout: 0.4 in classifier
- Epochs: ~17(early stop)

EfficientNet-B0

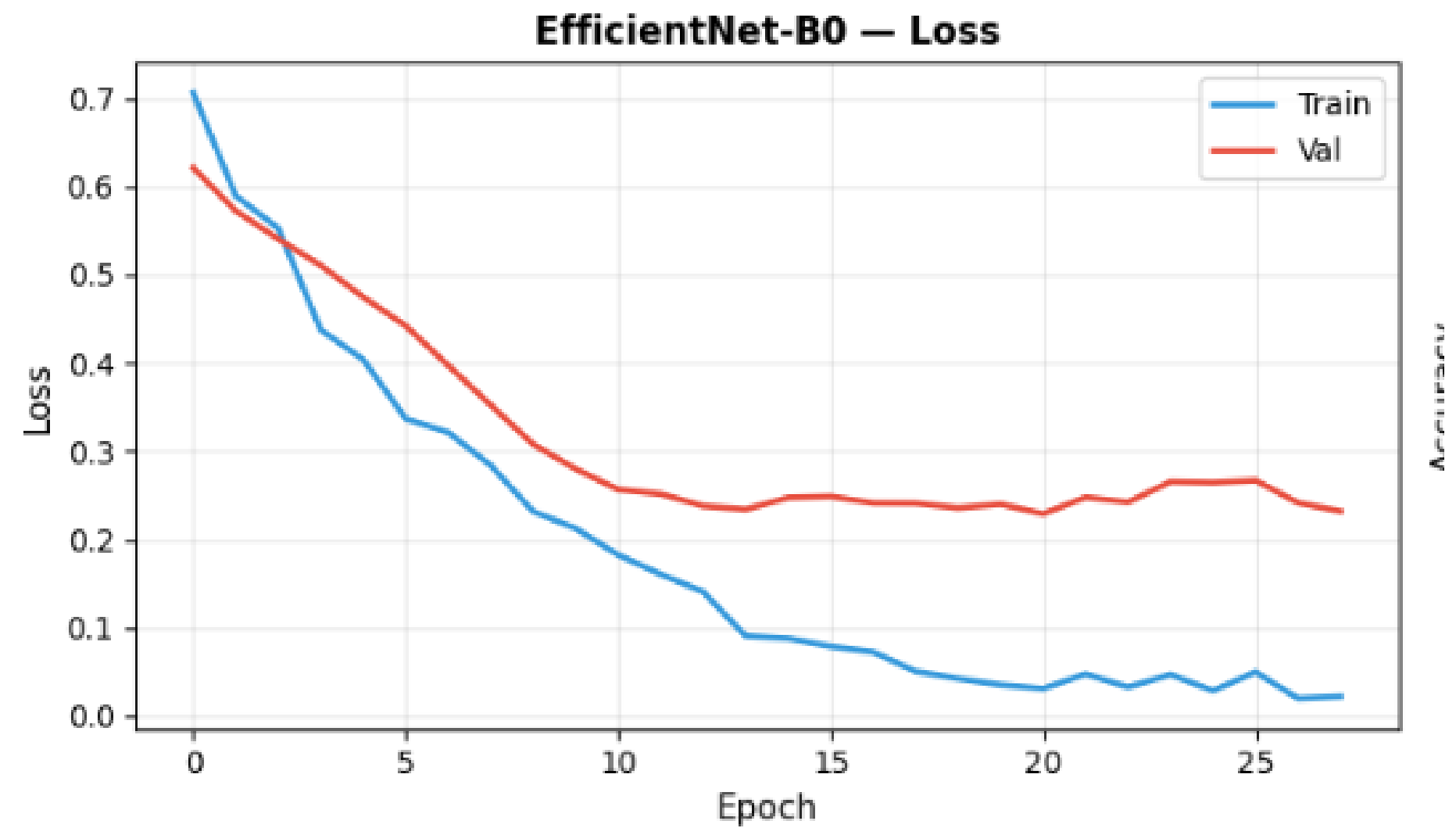
- Learning Rate: 1e-4 (lower)
- Dropout: 0.6 in classifier
- All layers trainable
- Epochs: ~28 (early stop)

Learning Curves — Training Progress

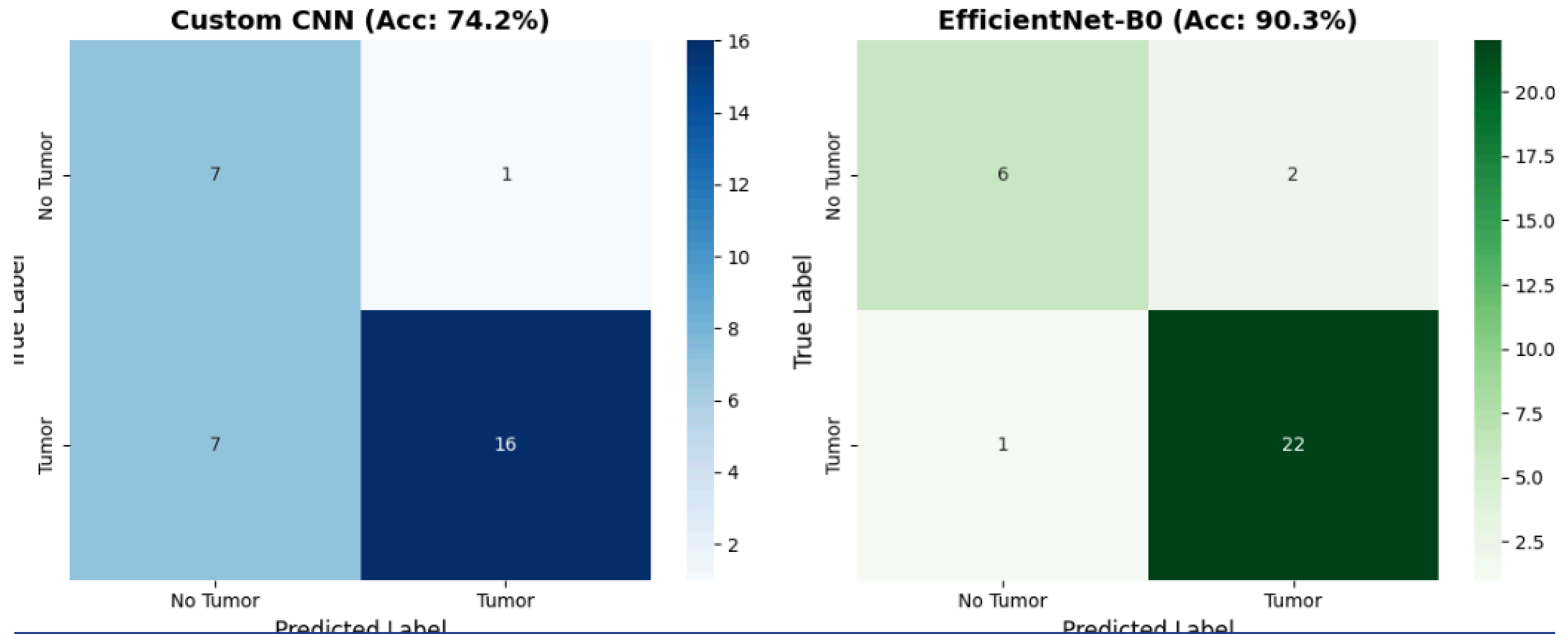
Custom CNN



EfficientNet-B0



Untangling The Confusion Matrix



NOTE: Substantial Improvement Detecting True Tumor Cases in EfficientNet; which means effectively reducing potential False Negative Misclassifications

False Negatives (FN) are most critical in medical AI — missing a tumor is far worse than a false alarm

Evaluation Results — Test Set Performance

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MODEL COMPARISON - Test Set Performance

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Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Custom CNN	0.741935	0.941176	0.695652	0.800000	0.896739
EfficientNet-B0	0.903226	0.916667	0.956522	0.93617	0.978261

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🏆 Best Model: EfficientNet-B0 (+16.1% better)

Key Findings

- ✓ EfficientNet achieved **90.3% accuracy** — excellent for 177 training images
- ✓ **High recall (95.6%)** — catches most tumors (critical for medical AI)
- ✓ **AUROC 0.978** — strong discriminative power
- ✓ **Transfer learning** outperforms custom CNN by 16% (90% vs 74%)

Explainable AI — Grad-CAM Visualization

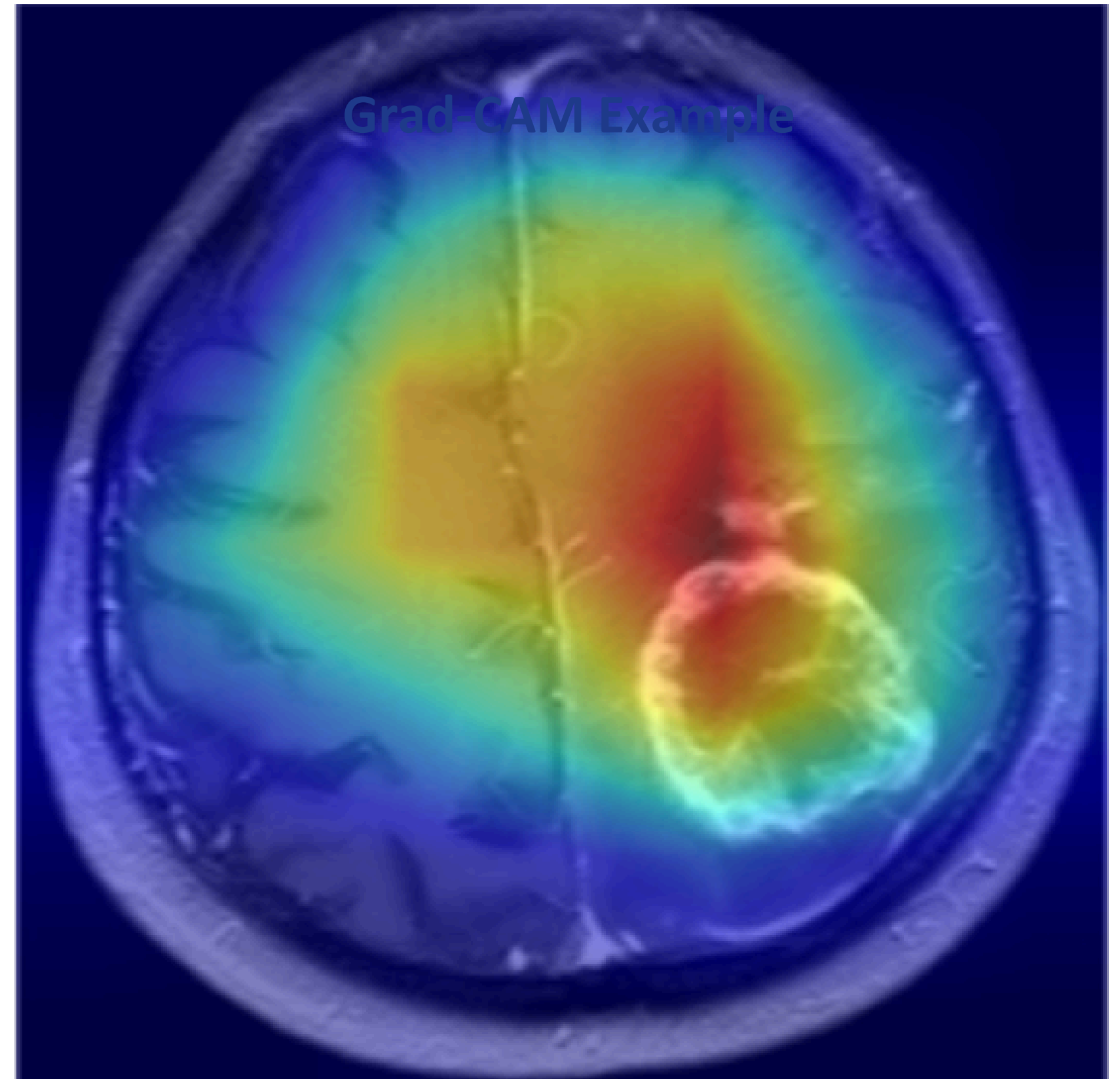
Why Explainability Matters

- Shows what the model "sees"
- Builds trust with doctors
- Identifies and mitigate biases
- Human-Understandable Visualization

Clinical Validation

- ✓ Model focuses on tumor regions
- ✓ Not learning from artifacts
- ✓ Explanations align with radiologist expectations

- Red/Orange Model thinks this is very important
- Yellow Somewhat important
- Blue Not important



Limitations & Challenges

Small Dataset

- Only 177 training images
- Limits model generalization
- Prone to overfitting

Class Imbalance

- 1.58:1 tumor:no-tumor
- Required class weighting
- Affects decision boundary

Binary Classification

- Only tumor vs no-tumor
- No tumor type/grade info
- No tumor size estimation

Risk Of Overfitting

- EfficientNet has millions of parameters
- Small datasets cannot fully constrain such models
- Data augmentation helps, but does not replace real diversity

Future Improvements & Next Steps



Expand Dataset

Collect 1000+ images from multiple hospitals and scanner types for better generalization



Multi-class Classification

Classify tumor types (glioma, meningioma, pituitary) and grade (benign vs malignant)



Advanced Models

Experiment with other advanced transfer models

Training Strategy Improvements

- **K- Fold Cross Validation**
- **Threshold Optimization**

Reflections & Lessons Learned

Technical Challenges

Overfitting was the biggest challenge — solved with heavy augmentation, dropout, and early stopping
Learning rate tuning was critical — too high caused loss spikes, too low caused stagnation
Custom CNN hit 76% ceiling — transfer learning was essential for 92% accuracy

Key Learnings

- Small datasets need careful handling
- Pretrained models >> custom CNNs
- Explainability is not optional
- Medical AI needs domain expertise
- Recall > Precision for screening

What I'd Do Differently

- Collect more data
- Add tumor segmentation from start (type of regularization to reduce overfitting)
- Involve radiologists in design

Key Takeaways

- ✓ Successfully built AI system for brain tumor detection with 92% accuracy
- ✓ Transfer learning (EfficientNet) significantly outperforms custom CNN
- ✓ Grad-CAM makes predictions explainable and clinically interpretable
 - ✓ Demonstrates potential for AI-assisted medical diagnosis